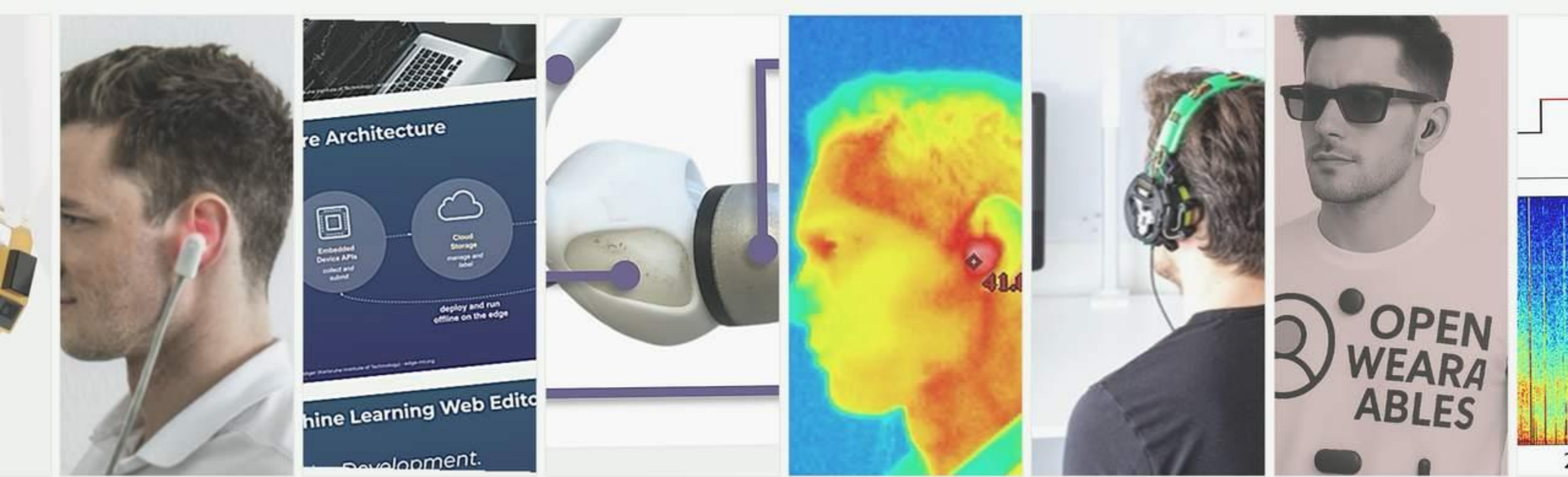




Human Computer Interaction SS26

Michael Beigl, TECO

Chapter

0

Organization

About Us



Lecturer

Prof. Dr. Michael Beigl

Head of the Chair for Pervasive
Computing Systems and the TECO
Research Group

Email: beigl@kit.edu

Tobias King

Exercise instructor for MoCloT

Email: king@teco.edu



Pervasive Computing

Translation: pervasive - “alles durchdringend”

IBM-Definition:

“Convenient access, through **a new class of appliances**, to relevant information with the ability to easily take action on it **when and where you need**”

Ubiquitous Computing

Ubiquitous = “allgegenwärtig”

Similar to Pervasive Computing

Ubicomp ISWC 2025





tecokit.edu

1

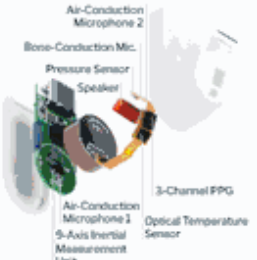
How to detect 25+ phenomena on the ears?

OpenEarable 2.0

Fully Open-Source AI Platform for Ear-Based Sensing Applications

IMWUT, Tuesday, Session B2 "Eyes and Ears", 14:15 - 15:45

Riddiger, T., Kitzner, M., Lepold, P., King, T., Mochizuki, G., Rapp, O., Parashar, A., Corke, C., & Biegt, M. (2025). OpenEarable 2.0: Open-source earphone platform for physiological ear sensing. In Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 9(1), 1-33.



Wearable HRF3140 - APPLICATION AND NETWORK Dual-Core Arm Cortex-M33

Analog Devices ADAU1860 - AUDIO DSP FastDSP Core & Tensilica HiFi2z DSP Core

Hardware and Software under MIT License

OPEN SOURCE

2

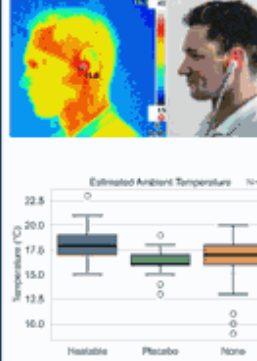
Feeling cold? Boost perceived temperature by 1.5°C with

Heatables

Infrared Light-Based Ear Heating Earphones

ISWC'25, Wednesday, Session D4 "Smart Devices", 10:45 - 12:15

Zhu, V., Kitzner, M., Hummel, L., Kitzner, M., Biegt, M., & Rapp, O. (2025). Heatables: Effects of Infrared LED-Induced Ear Heating on Thermal Perception, Comfort, and Cognitive Performance. In Proceedings of the 2025 ACM International Symposium on Wearable Computers.



Estimated Ambient Temperature 10-24

Temperature (°C)

Heatable Placebo None

3

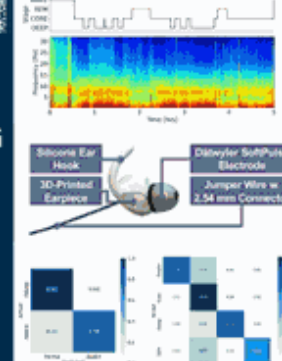
Track your sleep stages with in-ear electrodes using

OpenEarable ExG

Open-Source Platform for Biopotential-Sensing

EarComp '25, Workshop Monday, 9:00 - 17:45

Lepold, P., Lechler, J., Riddiger, T., & Biegt, M. (2025). Feasibility of In-Ear Single-Channel ExG for Wearable Sleep Monitoring in Real-World Settings. In Companion of the 2025 on ACM International Joint Conference on Pervasive and Ubiquitous Computing.



Silicone Ear Hook, 3D-Printed Earpiece, Disposable Sub-Pulse Electrode, Jumper Wires w/ 2.54 mm Connector

4

Dive into inaudible multi-channel sound zones powered by

UltrasonicSpheres

Using Off-the-Shelf Speakers and Earables

Ubicomp '25, Wednesday, Demo Session, 10:10 - 16:30

Kitzner, M., Zhu, V., Gertler, K., Biegt, M., & Riddiger, T. (2025). UltrasonicSpheres: Localized, Multi-Channel Sound Zones Using Off-the-Shelf Speakers and Earables. In Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing.



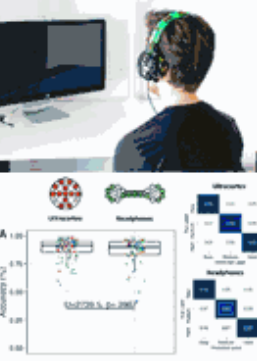
5

Can earphones predict how hard you are thinking?

Open ExG Headphones

Reliably Measure Cognitive Load via EEG Brain Data

Kitzner, M., T. Zang, C., Ivacki, G., & Riddiger, T. (2025). Advancing wearable EEG: Headphone EEG for cognitive load detection in lab and field. In Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 9(1), 1-26.



Accuracy (%)

Coding Meeting Planning Navigation

6

Get your multi-device study started in under five minutes with

HARNodes

An inexpensive open-source platform for inertial sensing

OpenWearables '25, Workshop Monday, Session 1, 9:55 - 10:05

Lepold, P., Riddiger, T., & Biegt, M. (2025). HARNodes: A Time-Synchronized, Open-Source, Multi-Device, Wearable System for Ad Hoc Field Studies. In Companion of the 2025 on ACM International Joint Conference on Pervasive and Ubiquitous Computing.



Head: 8, Chest: 9, Waist: 7, Wrists: 10, 11, Thighs: 5, 6, Shin: 3, 4, Feet: 1, 2

7

Can wearable haptic feedback make your rest more effective?

Wakeful Rest

Learn how and where haptics best support

ISWC'25, Wednesday, Session E2 "Health", 16:15 - 17:45

Lee, J., Fiebert, M., Lepold, P., Riddiger, T., & Biegt, M. (2025). Haptic Biofeedback for Wakeful Rest: Does Stimulation Location Make a Difference? In Proceedings of the 2025 ACM International Symposium on Wearable Computers.



Researched locations of vibration on the body: (A) Shoulder (B) Right to Left: Forearm, Wrist, Hand

Z-score normalized HANS SLEEP TREN LOWEST SLEEP TREN is interpreted as MORE PRESENT

8

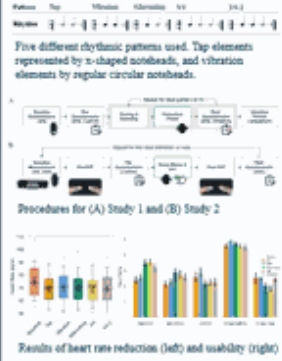
Feel calmer in minutes with haptic feedback

Smartwatch haptics for rest and sleep through

rhythmic, heart-synced vibration.

ISWC'25, Thursday, Session H3 "Fitness and Sports", 16:15 - 17:45

Asan, L., Dennis, M., Supraja, R., Tobin, R., Kai, K., & Michael, B. (2025). Closed-Loop Rhythmic Haptic Biofeedback via Smartwatch for Relaxation and Sleep Onset. In Proceedings of the 2025 ACM International Symposium on Wearable Computers.



Five different rhythmic patterns used. Tap elements represented by x-shaped noteheads, and vibration elements by regular circular noteheads.

Procedures for (A) Study 1 and (B) Study 2

Results of heart rate reduction (left) and usability (right)

9

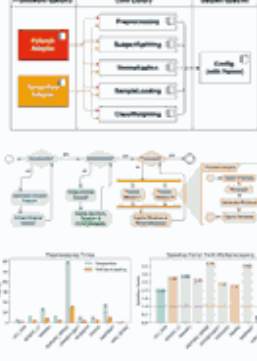
Boost research efficiency and reproducibility with

WHAR Datasets

A Data Handling Library for Wearable Human Activity Recognition

OpenWearables '25, Workshop Monday, Session 2, 11:35 - 11:45

Burns, M., King, T., Riddiger, T., Biegt, M., & Riddiger, T. (2025). WHAR Datasets: An Open-Source Library for Wearable Human Activity Recognition. In Companion of the 2025 on ACM International Joint Conference on Pervasive and Ubiquitous Computing.



Performance Time, Sensitivity for various activities

10

Are the fingers the next platform in wearable research?

Smart Rings

and their capabilities analyzed via systematic literature review

IMWUT, Wednesday, Session D4 "Smart Devices", 10:45 - 12:15

Wang, Z., Yu, R., Wang, X., Ding, J., Tang, J., Fang, J., He, Z., Li, Z., Riddiger, T., Xu, W., Zhang, X., Gao, H.-A., Gao, N., Yu, C., Shi, Y., & Wang, T. (2025). Computing with Smart Rings: A Systematic Literature Review. In Proceedings of the ACM on IMWUT '25.



Smart Rings capabilities: Activity Recognition, Health Monitoring, Navigation, Communication, etc.

11

Interactively experience your driving route with

RouteLLM

A Large Language Model with Native Route Context Understanding to Enable Context-Aware Reasoning

IMWUT, Wednesday, Session E3 "Vehicle Interaction", 16:15 - 17:45

Halgarten, P., Halbach, V., Koenig, E., Biegt, M., Grossen-Poppendorf, T. (2025). RouteLLM: A Large Language Model with Native Route Context Understanding to Enable Context-Aware Reasoning. In Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 9(1).



RouteLLM architecture: Contextual Reasoning, RouteLLM, RouteLLM, etc.

12

“Human-computer interaction is a discipline concerned with the **design, evaluation and implementation of interactive computing systems** for human use and with the study of major phenomena surrounding them.” (SIG CHI definition)

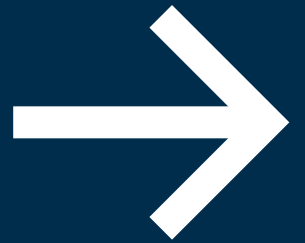
Integration in your study program



Stammmodul in Bachelor or Master Informatik (PO 2015, 22, 23),
Wahlmodul in Master Informatik
Wahlmodul in Master Informationswirtschaft
Advanced Mandatory Module Computer Science Master

6 ECTS Credits, 2+1 SWS Lecture and exercise

About this
lecture



Organization of the lecture

- Lecture Tuesday **8:10** – 9:30,
- (Exercise Wednesday 9:45 – 11:15)
with **Video** streaming and lecture translator
- Exercise start will be announced (depending on progress of lecture)
- Combination of lecture and exercise parts
 - Exercise are required to fully understand the concepts presented in the lecture
 - Exercise may include a presentation of your exercise results (depending on class size)

Mon	Tue	Wed	Thu	Fri
	Lecture			
		Lecture/ Exercise		

2 Hours per week lecture, every
second week exercise (2+1)

Exercise and Exam Information

Exercise:

- **Passing exercise is mandatory to register for the exam**
- some exercise are **mandatory** and must be submitted
- A passed exercise is eligible for any subsequent HCI lecture
- Exercise content can be used in the exam
- Exercise requires taking action e.g.
 - Interviews (form depends on Corona-situation)
 - Observation of people
 - Interview or hold workshop with peers
 - These are **basic skills** for HCI and thus mandatory



Exam:

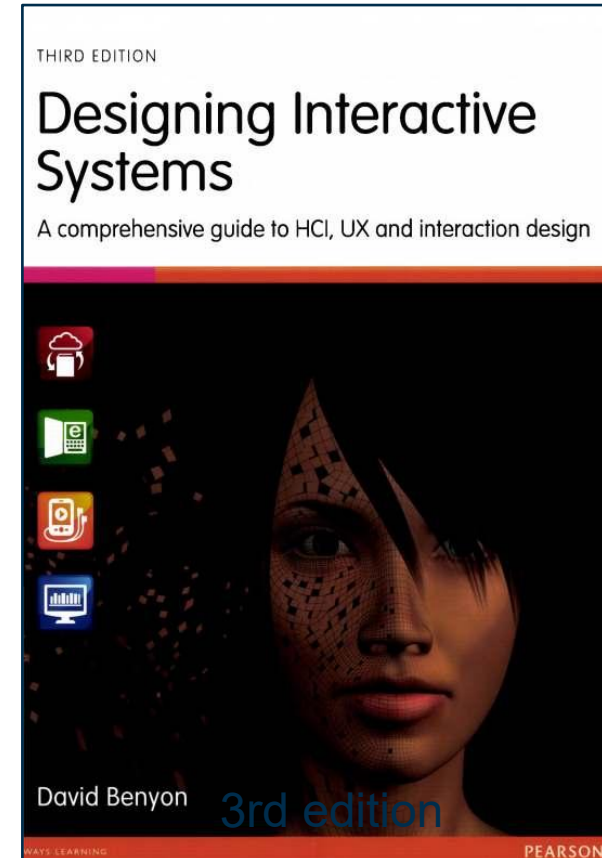
- Written exam every semester
- Next date to be annouced

Course material – Primary book reference

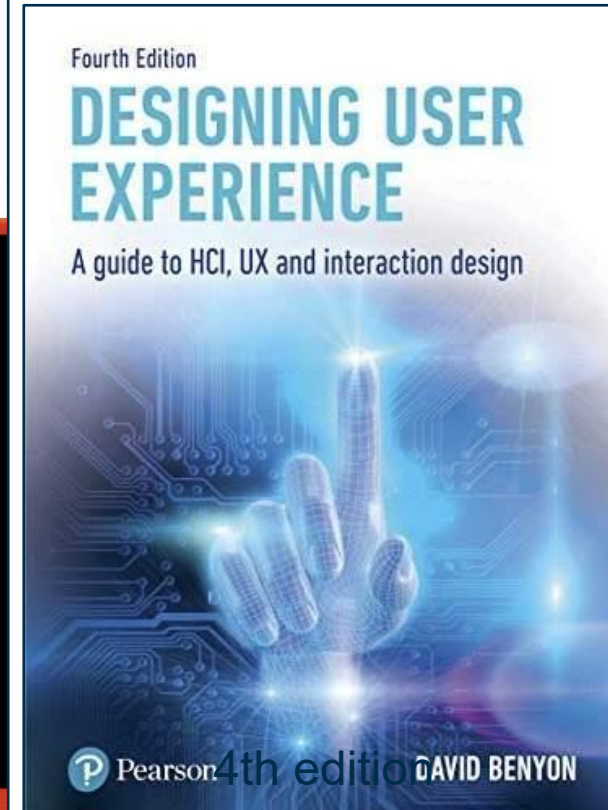


Designing Interactive Systems

- Author: David Benyon
- Benyon is a very good book with **timeless concepts** (Modern, easy to read and understand, useful as a **reference book** for projects)
- Anyone that is interested in this area should buy the book
- Available in library (Info, KIT)



► [image source](#)



► [image source](#)

Course material

Exercises are partially mandatory!

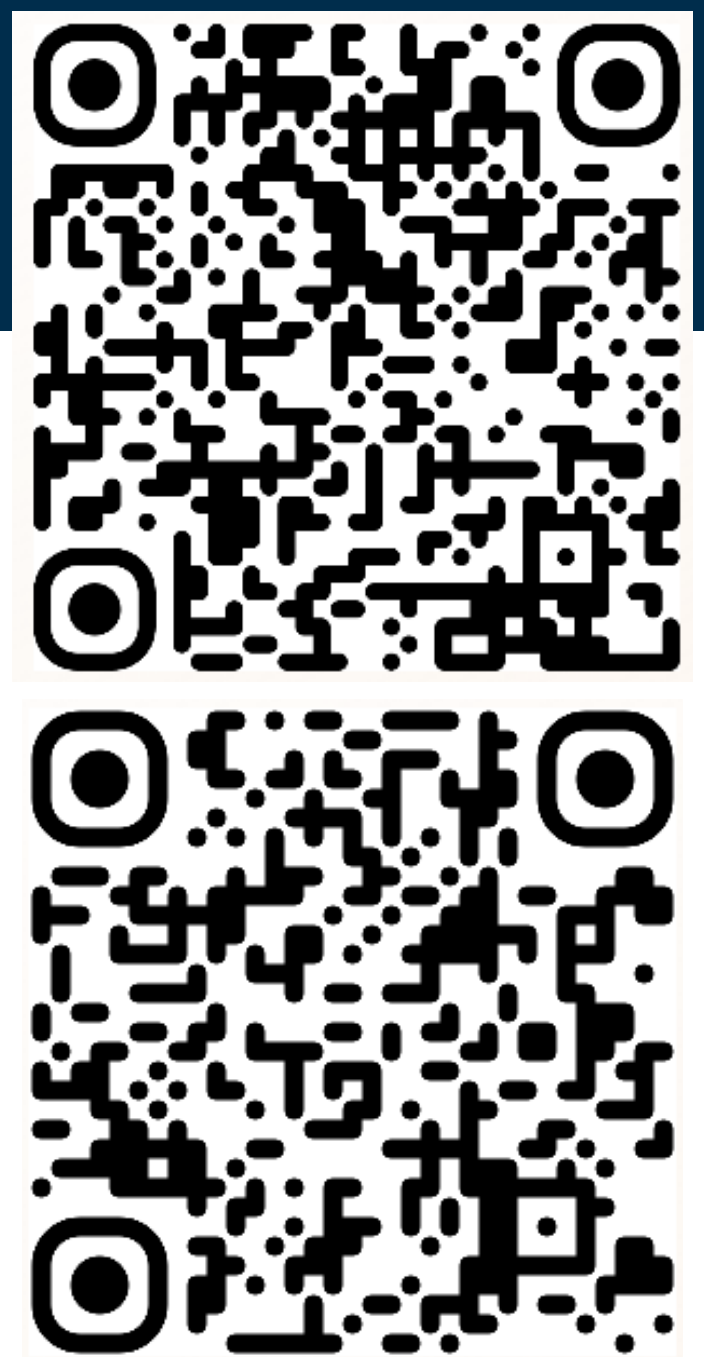
Link to **Exercise Material**

In ILIAS, because you submit in ILIAS!

<https://ilias.studium.kit.edu/goto.php/crs/2879639>

Lecture information

- Lecture slides, exam in ILIAS
- Some More Lecture information in <http://teco.kit.edu/education-hci>
- Some material is only accessible within the IP range of KIT → VPN



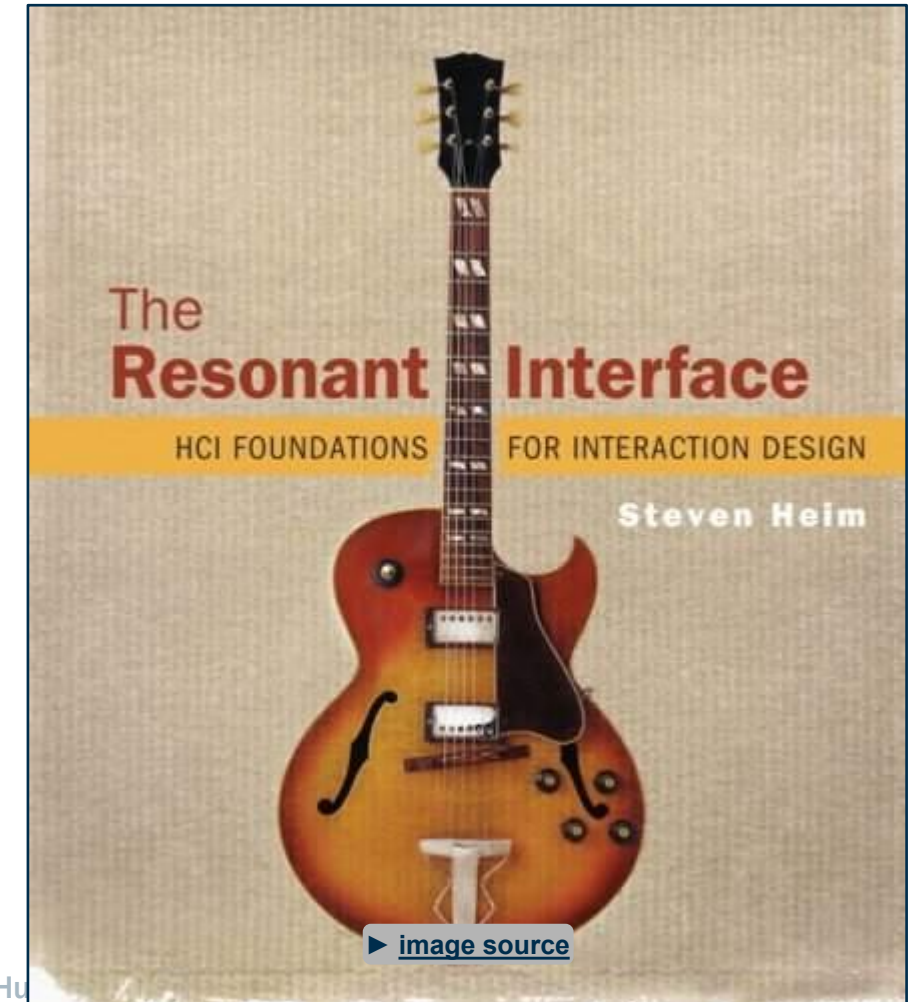
Course material – Secondary Sources



Sources given directly in the script

The Resonant Interface

- Author: Steven Heim
- ISBN 978-0-321-37596-4, Addison Wesley, 2008



Course material – Tertiary Sources

Engineering Psychology and Human Performance

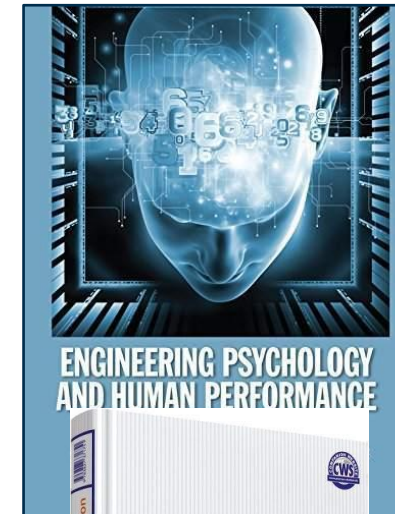
- Authors: Christopher Wickens and Justin Hollands
- ISBN: 0-312-04711-7, Prentice Hall, 1999

Human Factors in Engineering and Design

- Authors: Marc Sanders and Ernest McCormick
- ISBN: 0-071-12826-3, McGraw-Hill, 1993 (seventh edition)

Grundlagen der Mensch-Computer-Interaktion

- Author: Marcus Dahm
- ISBN 3-8273-7175-9, Pearson, 2006



► [image source](#)



► [image source](#)



► [image source](#)

Questions

Exam registration and other formal issues: sekretariat@tecو.edu

Exercise issues: king@tecو.edu

Lecture: After the lecture

Via E-Mail

- Lecture content only: Michael.Beigl@kit.edu
- You may have to send the E-Mail several times
- After one week non-reply your E-Mail is removed



Basic skills you will have after this lecture



- How do humans perceive their world
 - With their eyes and ears
- How do humans think
 - And what are limits of human thinking
- How to study humans behavior
 - Different ways to do so
 - How to set up this correctly, how to evaluate correctly
- How to analyse Human Computer Interaction
- How to design a Human Computer Interaction
- What that means in the age of AI

Lecture Content Overview



Basics of Cognitive Science

- Human Information Processing
- Perception

Formative Design and Analysis of Human Computer Interaction (HCI)

- Methods and Tools for Design, Analysis of Designs, PACT

Evaluation of HCI

- Basics of Human Studies and Statistics for Studies

Technology and Interaction in HCI

- Concepts, Models, Guidelines, Methods

Lecture Outline (preliminary)



Date / Day	Chapter	Topic
21 Apr 2026 (Tuesday)	Intro	Cognitive Foundations
22 Apr 2026 (Wednesday)	HIP	
28 Apr 2026 (Tuesday)	HIP	
29 Apr 2026 (Wednesday)	Exercise HIP / Lecture Integrated	
05 May 2026 (Tuesday)	Perception	
06 May 2026 (Wednesday)	Perception	
12 May 2026 (Tuesday)	Design Analysis	Design and Analysis
13 May 2026 (Wednesday)	Exercise Design Analysis and Perception	
19 May 2026 (Tuesday)	Design Analysis	
20 May 2026 (Wednesday)	Design	
26 May 2026 (Tuesday)	Pentecost	
27 May 2026 (Wednesday)	Pentecost	
02 Jun 2026 (Tuesday)	no lecture	
03 Jun 2026 (Wednesday)	Exercise Design	
09 Jun 2026 (Tuesday)	Design	
10 Jun 2026 (Wednesday)	Observation	Evaluation
16 Jun 2026 (Tuesday)	Observation	
17 Jun 2026 (Wednesday)	Exercise Observation	
23 Jun 2026 (Tuesday)	Interface and Interaction	Technology and Interaction
24 Jun 2026 (Wednesday)	Interfact and Interaction	
30 Jun 2026 (Tuesday)	Exercise Interface	

Lecture Content

- Human Capabilities stay the same
- Technology does not
- Large parts of the lecture content are dedicated to the human
- When technology changes, what we teach (and how) has to change as well
- ChatGPT (LLM, AgenticAI etc.) will substantially change many CS technology
- It will change HCI, and put an emphasis on it

